1. A theory based question based on space, time complexity and how to derive them and what are the terms and terminologies in DSA and what is algorithm and what is asymptotic notation and recursion and these might be asked along with examples too and the significance of these. Difference between ADT and Data Structures and difference between validating, analyzing and performing algorithms. Dictionary can be asked in any question but it is used for both BST and HASHING.

2. DLL or SLL – Anything can be asked but mostly would be programming side.

3. Stack or Queue – If given for 2 marks then just mention the application would be good enough and if it is asked for 5 marks or more then need to describe the step by step operation on how it will work. **CAN BE EITHER THEORY or PROGRAMMING**

4. SORTING AND SEARCHING, RECURSION – all the sorting techniques are included and time complexity is important as sorting would be asked based on the time complexity and for **QUICK SORT, MERGE AND HEAP SORT** even the time complexity needs to be derived if asked.

5. SORTING AND SEARCHING, RECURSION

6. Binary Search Tree (BST)

7. Graph

8. Hashing

9. Algorithm design techniques

10. Algorithm Design techniques

(6,7)* The terminologies in these are important and need to be prepared enough that we need to draw a tree or a graph and then explain that if they give the number of nodes and then ask how it traverses and all.

9,10 # These are only algorithms which are greedy method, BFS,DFS and all which have been taught in the last few classes including dynamic programming and N – Queens.

LAB

1. Linked List
2. Sorting Techniques
3. Graphs and Trees